Hostel Hive

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1.0	2026/03/19	Hasinthaka, Sanjuni, Thagya, Tasheen	Initial draft of the Software Requirement Specification.

Table 1: Version History

CHAPTER 1

INTRODUCTION

An introduction to the HostelHive project is provided in this chapter. Its background is presented, key problems associated with the current hostel management system are identified, and the aims and objectives of the proposed solution are outlined. The context and scope of the system are established, and a structured foundation is provided for the detailed requirements discussed in subsequent chapters.

The HostelHive system is proposed as a digital solution to improve the efficiency, security, and management of hostel operations at Bulugaha Girls' Hostel. The limitations of manual processes are addressed through the use of automated and centralized technologies, enabling improved monitoring and administration.

1.1 Project Background

Hostel management is considered a critical function within educational institutions, as the supervision of student accommodation, monitoring of student movements, and coordination of administrative activities are required. In many traditional hostel environments, including Bulugaha Girls' Hostel, these operations are carried out using manual, paper-based methods such as logbooks and physical records.

Manual processes are associated with several limitations. A significant amount of time is consumed in the recording and retrieval of information, and a high probability of human error is introduced. Student entry and exit tracking, room allocation, and leave management are performed manually, resulting in inefficiencies and an increased workload for wardens and administrative staff.

Furthermore, centralized data management is not available in the current system. As a result, inconsistencies in records may be introduced, and difficulties are encountered when historical data is retrieved. Real-time visibility into hostel activities is also not provided, by which the overall safety and monitoring of students may be negatively affected.

With advancements in digital technologies, opportunities for the modernization of hostel management systems are provided. Technologies such as QR code-based identification, real-time dashboards, and centralized databases are utilized to improve operational efficiency and data accuracy.

The HostelHive system is therefore proposed as an integrated solution in which core hostel management functionalities are combined into a single platform. Automated attendance tracking, digital room allocation, leave management, and communication features are incorporated to address the limitations identified in the existing system.

1.2 Problem Definition

The current hostel management system at Bulugaha Girls' Hostel is associated with several challenges by which operational efficiency and security are negatively impacted.

The key problems are identified as follows:

Manual and Time-Consuming Processes
Hostel activities are recorded using paper-based logbooks, by which operations are made time-consuming. Manual data entry is required, and an increased workload is experienced by administrative staff.

Lack of Real-Time Monitoring
Real-time information regarding student presence is not available. It is difficult for student presence to be determined by wardens at a given time, which may result in safety concerns.

Data Inaccuracy and Loss
Errors may be introduced during manual record-keeping, and data may be lost or damaged due to the use of physical documents. Reliable and consistent data management is therefore not ensured.

Inefficient Room Allocation Management
Room and bed allocation are managed manually, and accurate tracking of available resources is not maintained. As a result, inefficiencies are introduced, and misallocation of rooms may occur.

Lack of Structured Leave and Complaint Handling
A formal system for leave requests and maintenance complaints is not provided. As a result, delays and miscommunication may be introduced, and proper tracking of requests is not ensured.

Limited Communication Mechanisms
Announcements are not delivered through a centralized system. Important information may not be effectively communicated to all students, resulting in communication gaps.

These limitations indicate that a digital system is required in which hostel operations are automated, centralized, and efficiently managed.

1.3 Aims and Objectives

The primary aim of the HostelHive project is defined as the development of a centralized digital system through which hostel management operations can be efficiently controlled and monitored.

The following objectives are established to achieve this aim:

- A user-friendly system shall be developed in which hostel operations, including attendance tracking, room allocation, and student management, are digitized.
- A QR code-based entry and exit tracking mechanism shall be implemented, through which student movements are automatically recorded in real time.
- A centralized dashboard shall be provided, in which real-time attendance information is monitored by wardens.
- An efficient room and bed allocation system shall be implemented, through which hostel resources are accurately assigned and managed.
- An online leave management system shall be developed, through which leave requests are submitted by students and are approved or rejected by wardens.
- A complaint and maintenance management system shall be implemented, through which issues are reported and tracked systematically.
- A smart announcement system shall be introduced, through which communication between wardens and students is improved.

Hostel security shall be enhanced through controlled access, real-time monitoring, and role-based system usage.

Through the achievement of these objectives, a transition from manual hostel management to a modern, efficient, and secure digital system is expected to be achieved.

1.4 Purpose

The purpose of this Software Requirements Specification (SRS) document is defined as the provision of a detailed description of the requirements for the HostelHive system. This document is intended to be used as a reference for system development, validation, and future maintenance.

The HostelHive system is proposed as a replacement for the existing manual, paper-based hostel management system currently used at Bulugaha Girls' Hostel. In the existing system, key operations such as student entry and exit tracking, room and bed allocation, leave handling, and complaint management are recorded using physical logbooks and documents. As a result, inefficiencies are introduced, workload is increased, and a higher risk of data inaccuracies and loss is created.

Through this document, all functional and non-functional requirements of the proposed system are clearly defined. Hostel operations are intended to be digitized and automated, data accuracy is improved, security is enhanced, and real-time monitoring capabilities are provided.

This document is intended for the following stakeholders:

- System developers, by whom the system will be designed and implemented
- Project supervisors, by whom project progress and quality will be evaluated
- Hostel management staff, by whom the system will be used and validated
- Other relevant stakeholders involved in the system approval process

The requirements specified in this document shall be used as the basis for system design and development. Any modifications to these requirements shall be formally reviewed and approved prior to implementation.

1.5 Scope

The system under consideration is identified as HostelHive – Hostel Management System, by which hostel operations at Bulugaha Girls' Hostel are digitized and automated.

The scope of the system is defined by specifying the functionalities that are included (in-scope) and the limitations or exclusions (out-of-scope).

1.5.1 In-Scope

The following functionalities are included within the scope of the HostelHive system:

A digital identity and access management system shall be provided, through which users such as wardens, sub-wardens, security staff, and students are registered and authenticated.

- A student management module shall be implemented, in which student details such as personal information, contact details, and status are stored, updated,

and managed.

- A QR code-based entry and exit tracking system shall be introduced, through which student movements are automatically recorded using QR scanning devices.
- A real-time attendance dashboard shall be provided, through which students present inside or outside the hostel are monitored by wardens.
- A centralized room and bed allocation system shall be implemented, through which student accommodation is efficiently assigned and managed.
- A leave management system shall be provided, through which leave requests are submitted by students and are approved or rejected by wardens.
- A complaint and maintenance management system shall be implemented, through which issues are reported and their resolution is tracked.
- An announcement system shall be included, through which communication between wardens and students is facilitated.
- An in-app notification centre (inbox) shall be provided, through which users are informed about important events such as approvals, alerts, and announcements.

Basic reporting and analytics features shall be included, through which reports related to attendance, leave, and complaints are generated.

1.5.2 Out-of-Scope

The following aspects are explicitly excluded from the scope of the HostelHive system:

- Support for multiple hostels is not provided, and the system is designed only for Bulugaha Girls' Hostel.
- Financial and billing functionalities, including hostel fee management and payment processing, are not included.
- Advanced biometric authentication mechanisms, such as fingerprint or facial recognition, are not implemented.
- Offline system functionality is limited, and continuous internet connectivity is assumed.
- Integration with external third-party systems, such as university ERP platforms, is not included.
- Prevention of QR code duplication or misuse (e.g., the generation of fake QR codes by students) is not fully addressed within the current scope, and additional security mechanisms may be required in future enhancements.
- Device push notifications (e.g. FCM) are not implemented.

1.6 Definitions, Acronyms and abbreviations

The domain-specific terms, acronyms, and abbreviations used throughout this document are defined in this section to ensure a clear and consistent understanding of the system requirements.

1.6.1 Definitions

The following terms are defined for clarity:

HostelHive
A hostel management system that is designed to digitize and automate hostel operations at Bulugaha Girls' Hostel.

Student
An individual who resides in the hostel and uses the system for activities such as entry/exit tracking, leave requests, and complaint submission.

Warden
The primary administrative authority responsible for managing hostel operations, including student management, room allocation, and approvals.

Sub-Warden
An assisting authority who supports the warden in managing hostel activities.

Security Staff
Personnel responsible for monitoring hostel entry and exit activities and ensuring security compliance.

QR Code (Quick Response Code)
A machine-readable code assigned to each student, which is scanned to record entry and exit activities.

Attendance Dashboard
A system interface through which real-time information about student presence (inside or outside the hostel) is displayed.

Room Allocation
The process by which students are assigned to specific rooms and beds within the hostel.

Leave Request
A formal request submitted by a student to obtain permission to leave the hostel for a specified period.

Complaint Management
A system feature through which maintenance issues or concerns are reported and tracked.

Announcement
Information communicated by hostel management to students through the system.

1.6.2 Acronyms

The following acronyms are used in this document:

- SRS – Software Requirements Specification
- RBAC – Role-Based Access Control
- QR – Quick Response
- API – Application Programming Interface
- UI – User Interface
- DB – Database

1.6.3 Abbreviations

The following abbreviations are used:

- ESP32 – A microcontroller device used for QR code scanning
- HTTP – HyperText Transfer Protocol
- HTTPS – HyperText Transfer Protocol Secure
- CSV – Comma-Separated Values
- PDF – Portable Document Format

1.7 Document overview

This Software Requirements Specification (SRS) document is organized into six main chapters, each addressing key aspects of the HostelHive system. The structure is designed to provide a clear and systematic understanding of the system requirements, guiding the reader from an overview of the system to detailed specifications and analysis.

Each chapter is arranged to ensure that the requirements are presented in a logical and comprehensive manner, supporting effective communication among stakeholders, developers, and project supervisors.

Chapter 1: Introduction
An overview of the HostelHive system is provided in the Introduction chapter. The purpose of the document is defined, and the scope of the system is outlined. Key

definitions, acronyms, and abbreviations used throughout the document are presented to ensure clarity. The structure of the document is also described to guide the reader.

Chapter 2: Project Description
A high-level understanding of the system is presented in the Project Description chapter. The overall project scope is defined, and the stakeholders involved in the system are identified. Business rules governing the system are described, along with constraints that may affect system development. Assumptions made during requirement analysis are also specified.

Chapter 3: Specific Requirements
A detailed specification of system requirements is provided in this chapter. Functional requirements describing system behavior are documented in a structured format. Non-functional requirements defining system quality attributes such as performance, security, and usability are also specified. External interface requirements, including user interfaces, hardware components such as the QR scanning device, and communication protocols, are described. Additionally, data requirements related to system entities and storage are defined.

Chapter 4: System Models
Visual representations of the system are presented in the System Models chapter to illustrate its behavior and interactions. Use case diagrams are provided to show interactions between users and the system. Detailed use case descriptions are included to explain system functionalities. Activity diagrams are also presented to represent system workflows and processes.

Chapter 5: Project Analysis and Feasibility
The system is evaluated from an analytical perspective in this chapter. Alternative solutions to the identified problem are discussed and compared. A feasibility study is conducted to assess the technical, economic, operational, and schedule viability of the system. Potential risks associated with the system are identified, and mitigation strategies are proposed.

Chapter 6: References
All external sources, standards, and materials used in the preparation of this document are listed in this chapter, following the required referencing format.

Client Requirement Sign-off
A client requirement sign-off section is included at the end of the document. In this section, the client formally approves the system scope, requirements, and features. It is ensured that no further development is carried out without obtaining this approval.

CHAPTER 2

PROJECT DESCRIPTION

2.1 Project Scope

The Hostel Hive system is proposed as a specialized, IoT-integrated Hostel Management System developed exclusively for the Bulugaha Girls' Hostel at the University of Kelaniya. Currently, manual, paper-based logbooks are relied upon by the hostel for tracking student movement, by which high inefficiency and susceptibility to human error are introduced, and emergency auditing is made nearly impossible. The entire digitization of these administrative workflows is defined as the scope of this project, through which security and operational efficiency are enhanced.

Automated attendance tracking via ESP32 hardware, centralized room and bed allocation management, and a digital leave approval process will be provided by the system. A Modular Monolithic architecture is utilized by the system, and Domain-Driven Design (DDD) is employed so that the core domains of identity management, attendance logging, and administration are cleanly separated.

Explicit Exclusions: The system is strictly bounded as a single-client operational tool. Multi-tenant Software-as-a-Service (SaaS) architectures, cross-university scalability, and integrated financial billing modules are explicitly excluded, ensuring that the development focus is maintained purely on solving the immediate operational bottlenecks of the Bulugaha Girls' Hostel.

2.2 Stakeholders

The stakeholders of the Hostel Hive system are identified as the individuals and groups by whom the system is directly or indirectly affected, or by whom an interest in its development, deployment, and operation is possessed. The key stakeholders identified for the system are described below:

Client / Hostel Management: The ultimate business needs and institutional authority are represented by Dr. Kasun Fernando and the Student Welfare Division. The system is utilized by this group to improve operational efficiency, enhance hostel security, and support data-driven decision-making processes regarding resource allocation.

Students: Students are considered the primary end-users of the system. The mobile application and hardware endpoints are utilized by students to perform activities such as scanning QR codes for entry and exit, submitting digital leave requests, viewing global announcements, and reporting maintenance complaints. Accurate tracking of their attendance is ensured through their interaction with the ESP32 hardware.

Warden: The warden is designated as the main administrative authority by whom daily hostel operations are managed. The web dashboard is used by the warden so that real-time student attendance is monitored, leave requests are approved or rejected, room and bed allocations are managed, complaints are reviewed, and announcements are broadcasted.

Sub-Warden: The warden is supported by the sub-warden in managing hostel activities. System functionalities similar to those of the warden may be accessed by the sub-warden, depending on assigned Role-Based Access Control (RBAC) permissions. Assistance in monitoring and administrative tasks is provided through the system.

Security Staff: The physical monitoring of the hostel entry and exit points is assigned to the security staff. The system dashboard is used by security personnel to ensure that proper entry and exit procedures are followed by students and that successful QR code scans at the hardware terminals are verified.

System Administrator: System configurations, user accounts, and overall system maintenance are managed by the system administrator. Tasks such as user activation/deactivation, system monitoring, and issue handling are performed by the administrator to ensure continuous and secure system operation.

Developers: The design, implementation, and maintenance of the Hostel Hive system are designated as the responsibility of the software engineering undergraduate team. The requirements specified in this document are used by the developers as the strict reference architecture for system development and hardware integration.

Project Supervisor: The development of the system is overseen by the project supervisor from the Software Engineering Teaching Unit. The final system deliverables and this SRS document are evaluated by the supervisor based on the academic objectives defined for the SENG 31242 course.

2.3 Business Rules

The system must strictly enforce the following operational business rules to maintain security and order within the hostel:

- **Strict Access Control:** Only actively registered students with valid, system-generated QR codes are permitted to log entry and exit times. If a student's university registration expires or they vacate the hostel, the system must instantly invalidate their QR token.
- **Leave Authorization Hierarchy:** All extended student leave requests (e.g., weekend travel) must be formally approved by a registered Hostel Warden through the system dashboard. Students cannot self-approve leave.
- **Capacity Constraints:** Overbooking must be strictly prevented by the system. A single bed can only be assigned to one active student at any given time, and the predefined maximum bed count for any specific room cannot be exceeded by wardens.
- **Hardware Syncing Rule:** If Wi-Fi connectivity is lost by the ESP32 scanners, the scanned QR data must be securely queued locally, and the attendance logs must be automatically pushed to the central PostgreSQL database immediately upon network restoration.

2.4 Constraints

The architecture and operation of the Hostel Hive system are bounded by the following technical and environmental constraints:

- **Hardware Constraints:** The physical entry and exit tracking mechanism is constrained to ESP32 microcontrollers. These devices have limited processing power and memory, requiring highly optimized C++ code for rapid QR scanning and payload transmission.
- **Technological Constraints:** The project stack is strictly defined. The backend must be developed using Go to handle concurrent IoT requests efficiently. Data must be stored in a PostgreSQL relational database, with Firebase handling real-time state changes. The administrative web dashboard must be built using React, while the student-facing mobile application utilizes Flutter.
- **Environmental Constraints:** The IoT scanning endpoints are entirely dependent on a continuous power supply and a stable Wi-Fi connection within the hostel lobby to transmit real-time data to the cloud infrastructure.
- **Schedule Constraints:** The complete Software Requirement Specification (SRS) and the officially signed Client Agreement Letter must be submitted on or before the 22nd of March 2026 to meet the SENG 31242 course deadlines.
- **Notification Constraint:** Push notifications are not supported.

2.5 Assumptions

The functional and non-functional requirements detailed in this document are based on the following foundational assumptions:

- An accurate, digitized initial CSV dataset of all currently registered students and the exact room layout will be provided by the Student Welfare Division so that the database can be populated prior to system deployment.
- The necessary Wi-Fi infrastructure within the hostel will be maintained by the university IT department to ensure the ESP32 modules remain connected to the backend servers.
- Smartphones capable of clearly displaying the Flutter mobile application are possessed by students, ensuring the screen brightness and clarity are sufficient for the QR code to be read by the physical ESP32 scanner without manual warden intervention.
- Basic digital literacy is possessed by the wardens, and a brief training session will be provided to them so that effective navigation of the React web dashboard is understood.

CHAPTER 3

SPECIFIC REQUIREMENTS

3.1 Functional Requirements

All functional requirements of the HostelHive system are defined below. Each requirement is assigned a unique ID, priority level, and source stakeholder.

ID	Description	Priority	Source	Status
FR001	Registration of accounts shall be allowed for users (warden, sub-warden, student).	High	Admin	Proposed
FR002	Secure login functionality using authentication mechanisms shall be provided.	High	All Users	Proposed
FR003	Role-based access control (RBAC) shall be enforced.	High	Admin	Proposed
FR004	Passwords shall be allowed to be reset and changed by users.	Medium	All Users	Proposed
FR005	User accounts shall be allowed to be activated or deactivated by the administrator.	Medium	Admin	Proposed
FR006	Student records shall be allowed to be added, updated, and deleted.	High	Warden	Proposed
FR007	Student details (ID, name, contact, guardian details) shall be stored.	High	Warden	Proposed
FR008	The uploading of student profile images shall be supported.	Low	Warden	Proposed
FR009	Searching and filtering of student records shall be allowed.	Medium	Warden	Proposed
FR010	A unique, scannable QR code shall be generated for each student.	High	Warden	Proposed
FR011	Entry and exit scans shall be recorded strictly via ESP32 hardware endpoints.	High	Admin	Proposed

FR012	Exact timestamps for each entry and exit event shall be logged.	High	Warden	Proposed
FR013	A student's IN and OUT status shall be automatically toggled upon scanning.	High	Warden	Proposed
FR014	Duplicate QR scans within a defined cooldown period shall be prevented.	Medium	Admin	Proposed
FR015	Real-time synchronization between ESP32 nodes and the backend shall be provided.	High	Admin	Proposed
FR016	Real-time attendance counts (inside/outside) shall be displayed on the dashboard.	High	Warden	Proposed
FR017	Historical attendance summaries and analytical counts shall be provided.	High	Warden	Proposed
FR018	Available rooms, blocks, and beds shall be allowed to be viewed by wardens.	High	Warden	Proposed
FR019	Individual students shall be allowed to be assigned to specific beds by wardens.	High	Warden	Proposed
FR020	Existing room allocations shall be allowed to be updated or revoked.	Medium	Warden	Proposed
FR021	Extended leave requests shall be allowed to be digitally submitted by students.	High	Student	Proposed
FR022	Pending leave requests shall be allowed to be approved or rejected by wardens.	High	Warden	Proposed
FR023	A permanent log of all historical leave records shall be maintained.	Medium	Warden	Proposed
FR024	Students shall be automatically marked as OUT during the approved leave duration.	High	Admin	Proposed
FR025	Maintenance or disciplinary complaints shall be allowed to be submitted by students.	High	Student	Proposed
FR026	Active complaints shall be allowed to be viewed, filtered, and managed by wardens.	High	Warden	Proposed
FR027	Complaints shall be allowed to be digitally marked as resolved.	Medium	Warden	Proposed
FR028	Global announcements shall be allowed to be broadcasted by wardens.	High	Warden	Proposed

FR029	Announcements shall be delivered to the student mobile application.	High	Student	Proposed
FR030	Comprehensive reports on attendance, leave, and complaints shall be generated.	Medium	Admin	Proposed
FR031	Generated reports shall be exported in PDF and Excel formats.	Low	Admin	Proposed
FR032	In-app notifications for leave approvals and critical updates shall be shown in the notification centre (inbox) of the student mobile application.	Medium	Student	Proposed

Table 2: Functional Requirements

3.2 Non-Functional Requirements

The non-functional requirements of the HostelHive system are categorized based on ISO/IEC 25010 standards.

3.2.1 Performance Requirements

ID	Requirement	Acceptance Criteria
NFR001	User requests shall be responded to quickly.	95% of queries shall be processed within 2 seconds under normal load.
NFR002	Multiple users shall be handled simultaneously.	The system shall support at least 100 concurrent users without performance degradation.

Table 3.1: Performance Requirements

3.2.2 Reliability Requirements

ID	Requirement	Acceptance Criteria
NFR003	Continuous operation shall be ensured.	System uptime shall be at least 99% per month.
NFR004	Failures shall be recovered from gracefully.	The system shall recover within 5 minutes after a critical failure.

Table 3.2: Reliability Requirements

3.2.3 Usability Requirements

ID	Requirement	Acceptance Criteria
NFR005	A user-friendly experience shall be provided for non-technical staff.	90% of wardens shall be able to perform room allocations without external training.
NFR006	A highly navigable interface shall be provided.	All major administrative actions shall be accessible within 3 clicks from the dashboard.

Table 3.3: Usability Requirements

3.2.4 Security requirements

ID	Requirement	Acceptance Criteria
NFR007	Secure data communication shall be ensured.	All data in transit shall be encrypted using TLS 1.2 / HTTPS.
NFR008	Authentication shall be strictly enforced.	Unauthorized access attempts to administrative endpoints shall be blocked 100% of the time.
NFR009	All user inputs shall be validated.	100% of inputs shall be sanitized to prevent SQL injection and XSS attacks.

Table 3.4: Security Requirements

3.2.5 Scalability requirements

ID	Requirement	Acceptance Criteria
NFR010	Continuous data growth shall be supported by the database.	The PostgreSQL database shall support storing up to 5 years of attendance logs without manual archiving.

Table 3.5: Scalability Requirements

3.2.6 Maintainability requirements

ID	Requirement	Acceptance Criteria
NFR011	Easy maintenance and patching shall be facilitated.	Minor code updates shall be deployable within a 2-hour maintenance window.

NFR012	A modular architecture shall be supported.	Core domains (e.g., Attendance vs. Leave) shall be independently updatable.

Table 3.6: Maintainability Requirements

3.2.7 Portability requirements

ID	Requirement	Acceptance Criteria
NFR013	Modern digital platforms shall be supported for execution.	The web dashboard shall support Chrome/Firefox, and the mobile app shall support Android 10+.
NFR014	Easy deployment across environments shall be facilitated for the backend.	The Go backend and PostgreSQL database shall be containerized using Docker for deployment flexibility.

Table 3.7: Portability Requirements

3.3 External Interface Requirements

The interactions of the HostelHive system with users, hardware devices, external software systems, and communication networks are defined by the external interface requirements. Given the importance of these interfaces, they are designed so that the ability for seamless integration with users is provided, and the means for ensuring that the overall functioning of the HostelHive system is operated efficiently and effectively are established.

3.3.1 User Interface Requirements

User Interfaces for the HostelHive System are designed to be simple, intuitive, and user-friendly; therefore, an easy way is provided for any user, whether a student, warden, or security personnel, to interact with the HostelHive system.

- A graphical user interface (GUI) shall be provided for both web and mobile platforms.
- The interface shall be made responsive and adaptable to different screen sizes (mobile, tablet, desktop).
- Navigation shall be kept simple and consistent across all modules.
- All major functionalities shall be made accessible within minimal steps.
- Clear labels, icons, and instructions shall be provided for user actions.
- Validation and error messages shall be included in input fields.

- Basic accessibility features such as readable fonts and proper contrast shall be supported by the system.
- Role-based dashboards shall be displayed according to user type (warden, student, etc.).

3.3.2 Hardware Interface Requirements

Interactions with specific hardware components shall be performed by the HostelHive system to support its core functionality.

- An interface with an ESP32 microcontroller device shall be established by the system for QR code scanning.
- A QR code scanner module (camera-based) shall be connected to the ESP32 device.
- QR codes shall be read and data shall be transmitted to the backend system by the ESP32 device.
- Standard computing devices such as desktops, laptops, and smartphones shall be supported by the system for user access.
- Internet connectivity shall be required for communication between hardware devices and the system.

3.3.3 Software Interface Requirements

The system shall integrate with various software components and services to ensure proper functionality.

- A backend server shall be implemented using Go (Golang) for handling business logic.
- A database management system (PostgreSQL) shall be used for data storage and retrieval.
- Frontend applications shall be developed using Flutter (mobile) and React (web).
- APIs shall be used to enable communication between frontend, backend, and hardware components.
- Authentication mechanisms (e.g., JWT-based authentication) shall be implemented for secure access.
- Notifications shall be shown in the in-app notification centre; no push service is used.

3.3.4 Communication Interface Requirements

The manner in which data is exchanged between system components is defined by the communication interface.

- Communication between frontend, backend, and hardware devices shall be performed over the internet.
- HTTP/HTTPS protocols shall be used by the system for secure data transmission.
- Data exchanged between components shall be formatted using JSON.
- Communication with the backend shall be performed by the ESP32 device via RESTful APIs.
- Real-time updates shall be supported through efficient data synchronization mechanisms.
- Secure communication shall be ensured through encryption protocols.

3.4 Data Requirements

The structure, storage, management, and integrity of data used within the system are defined by the data requirements of the HostelHive system. All data shall be stored in a centralized database to ensure consistency, reliability, and efficient retrieval.

3.4.1 Data Entities

The system shall manage several core data entities. Each entity represents a key component of the hostel management process.

User
Information related to system users (warden, sub-warden, security staff, students) shall be stored. Attributes include user ID, name, role, email, password, and account status.

Student
Student-specific details shall be maintained, including student ID, name, contact information, guardian details, and profile image.

QR Code
Each student shall be associated with a unique QR code used for identification during entry and exit.

Attendance Record
Entry and exit data shall be recorded with timestamps. Each record shall include student ID, date, time, and status (IN/OUT).

Room
Room details such as room number, capacity, and availability status shall be stored.

Bed Allocation
Information regarding student room and bed assignments shall be maintained.

Leave Request
Details such as student ID, leave dates, reason, approval status, and warden comments shall be included in the leave application records.

Complaint
Details such as student ID, description, submission date, status, and resolution information shall be included in the complaint records.

Announcement
Announcements created by wardens shall be stored, including title, message content, date, and target audience.

Notification
Notifications sent to users shall be recorded. Data such as type, message, recipient, timestamp, and read status shall be included.

3.4.2 Data Storage Requirements

- All system data shall be stored in a PostgreSQL relational database.
- Data shall be organized into structured tables corresponding to each entity.
- Primary keys shall be used to uniquely identify each record.
- Foreign keys shall be used to maintain relationships between entities.
- Data backup mechanisms shall be implemented to prevent data loss.

3.4.3 Data Integrity and Validation

- Data accuracy shall be ensured through input validation at both frontend and backend levels.
- Mandatory fields shall be enforced for critical data entries.
- Duplicate records shall be prevented using unique constraints.
- Referential integrity shall be maintained between related entities.
- Data consistency shall be ensured across all system modules.

3.4.4 Data Security Requirements

- Sensitive data such as passwords shall be stored in encrypted form.
- Access to data shall be controlled using role-based permissions.

- Secure communication protocols (HTTPS) shall be used for data transmission.
- Unauthorized access to data shall be restricted.

3.4.5 Data Retention and Backup

- System data shall be retained for a defined period as required by hostel policies.
- Regular database backups shall be performed to ensure data recovery in case of failure.
- Backup data shall be stored securely to prevent unauthorized access.

3.4.6 Data Relationships (Overview)

The relationships between key entities shall be maintained as follows:

- A student or staff role may be mapped to a user account.
- One unique QR code shall be possessed by a student.
- Multiple attendance records may be linked to a student.
- Multiple leave requests and complaints may be submitted by a student.
- A student shall be assigned to one room/bed at a time.
- Multiple announcements may be created and multiple records may be managed by a warden.

CHAPTER 4

SYSTEM MODELS

4.1 Use Case Diagram

[Figure 1: Use Case Diagram – HostelHive System actors (ESP32 Device System, Student, Security, Sub-Warden, Warden, Admin) and use cases, as presented in the source document.]

4.2 Use Case Descriptions

Field	**Description**
Use Case ID	UC001
Use Case Name	Login /Authentication
Description	The process through which registered users' identities are authenticated to access the HostelHive system is described by this case. Credentials are provided by the users and redirection to respective role-based dashboards is performed by the system.
Actor	Student, Warden, Sub-Warden, Admin
Pre-Condition	A registered account in the system must be possessed by the user. The login page must be navigated to by the user.
Post-Condition	The user is authenticated and access to the system is granted. The user is redirected to their role-based dashboard by the system.
Main Flow	1. The HostelHive login page is navigated to by the user. 2. The login form requesting email/student ID and password is displayed by the system. 3. Credentials are entered and the form is submitted by the user. 4. The credentials are validated against stored records by the system. 5. If verified, the user is logged in and redirected to their role-based dashboard by the system.
Alternative Flow	If invalid credentials are entered, an error message is displayed by the system. If the user's account has been deactivated, the user is notified to contact the administrator by the system.

	If the password is forgotten, the Reset Password feature may be used by the user.

Table 4: Use-Case Description 1

Field	**Description**
Use case ID	UC002
Use Case Name	Reset Password
Description	Login/Authentication is extended by this case. The process by which a forgotten password is reset or an existing password is changed by the user is outlined.
Actor	Student, Warden, Sub-Warden, Admin
Pre-Condition	For password reset: a registered email address must be possessed by the user. For password change: the system must be logged into by the user.
Post-Condition	The user's password is updated in the database. The new password can be used by the user to log in.
Main Flow	1. "Forgot Password" on the login page is clicked by the user. 2. The user is prompted to enter their registered email by the system. 3. A password reset link is sent to the email by the system. 4. The link is clicked and a new password is entered by the user. 5. Password complexity is validated and the new password is saved by the system. 6. The reset is confirmed and the user is redirected to the login page by the system.
Alternative Flow	If the entered email is not registered, an error message is displayed by the system. If security requirements are not met by the new password, a stronger password is requested by the system. If the reset link has expired, a new link is prompted to be requested by the system.

Table 5: Use-Case Description 2

Field	**Description**
Use Case ID	UC003
Use Case Name	Manage Student Profiles

Description	Student profiles within the system are allowed to be added, updated, searched, and deleted by the admin through this case. Student information including personal details, contact information, and guardian details is managed through this module.
Actor	Admin
Pre-Condition	The system must be logged into by the admin. Student data must be available for entry.
Post-Condition	Student profiles are created, updated, or deleted in the database. Changes are reflected immediately across the system.
Main Flow	1. The student profile management module is navigated to by the admin. 2. All registered students with search and filter options are displayed by the system. 3. A student profile is selected to be added, updated, or deleted by the admin. 4. Add: Student details (ID, name, faculty, contact, guardian info) are entered and submitted by the admin. 5. Update: An existing profile is selected, required fields are modified, and changes are saved by the admin. 6. Delete: A student profile is selected and deletion is confirmed by the admin. 7. The data is validated and changes are committed by the system.
Alternative Flow	If the student ID already exists, a duplicate error is displayed by the system. If mandatory fields are left empty, completion is prompted by the system. For bulk registration, the Bulk Import feature may be used by the admin.

Table 6: Use-Case Description 3

Field	**Description**
Use Case ID	UC004
Use Case Name	Bulk Import Students (CSV)
Description	Manage Student Profiles is extended by this case. Multiple student records are allowed to be imported at once by the admin through uploading a CSV file obtained from the Student Welfare Division.
Actor	Admin
Pre-Condition	The system must be logged into by the admin. A properly formatted CSV file with student data must be available.

Post-Condition	All valid student records from the CSV file are created in the system. Invalid records are flagged in an error report.
Main Flow	1. The student profile module is navigated to and "Bulk Import" is selected by the admin. 2. The CSV upload interface with format guidelines is displayed by the system. 3. The admin uploads the CSV file. 4. Each record in the file is parsed and validated by the system. 5. Valid records are imported and student accounts are created by the system. 6. A summary showing successful imports and any errors is displayed by the system.
Alternative Flow	If the CSV file format is invalid, the upload is rejected and format requirements are displayed by the system. If duplicate student IDs are found, they are flagged and those entries are skipped by the system. If the file is empty or corrupted, the admin is notified by the system.

Table 7: Use-Case Description 4

Field	**Description**
Use Case ID	UC005
Use Case Name	Scan QR Code
Description	The process where the university ID card QR code is presented by a student to the ESP32 scanner device at the hostel gate is described by this case. Verify Student Identity and Log Entry/Exit are included as mandatory sub-processes.
Actor	Student (scanning is initiated), ESP32 Device System (the scan is processed)
Pre-Condition	A university ID card with a valid QR code must be possessed by the student. The ESP32 device must be powered on and connected to the network.
Post-Condition	The QR code data is read and transmitted to the backend server for verification and logging.
Main Flow	1. The hostel gate is approached and the university ID card QR code is presented to the ESP32 scanner by the student. 2. The QR code data is read by the ESP32 device.

	3. The scanned data is transmitted to the backend server by the device. 4. Verify Student Identity by the system. 5. Upon successful verification, Log Entry/Exit proceeded to the system.
Alternative Flow	If the QR code is damaged or unreadable, decoding is failed by the ESP32 device and no data is transmitted. If the ESP32 device is offline, scan data is cached locally and pushed once connectivity is restored.

Table 8: Use-Case Description 5

Field	**Description**
Use Case ID	UC006
Use Case Name	Verify Student Identity
Description	This case is included in Scan QR Code. The process of verifying the scanned student's identity and account status against the system database is described.
Actor	ESP32 Device System (triggered), System (processed)
Pre-Condition	The QR code data must have been successfully scanned and received by the backend server.
Post-Condition	The student's identity is confirmed or rejected. The verification result is returned to the calling process.
Main Flow	1. The QR code data is received from the ESP32 device by the system. 2. The student ID is extracted from the QR data by the system. 3. The database is queried to find the matching student record by the system. 4. Whether the student's account is active is checked by the system. 5. If the student is verified and active, a successful verification is returned by the system.
Alternative Flow	If no record is matched by the student ID, the scan is rejected and the failed attempt is logged by the system. If the student's account is deactivated, entry is rejected and the attempt is logged as unauthorized by the system.

Table 9: Use-Case Description 6

Field	**Description**
Use Case ID	UC007

Use Case Name	Log Entry/Exit
Description	This case is included in Scan QR Code. The automatic recording of a student's entry or exit with a timestamp once their identity is verified is described.
Actor	System (automatic process)
Pre-Condition	The student's identity must have been successfully verified.
Post-Condition	The entry or exit is recorded with an exact timestamp. The student's IN/OUT status is toggled. The attendance dashboard is updated in real time.
Main Flow	1. The student's current status (IN or OUT) is checked by the system. 2. The status is automatically toggled (IN → OUT or OUT → IN) by the system. 3. The event is recorded with the current timestamp by the system. 4. The attendance dashboard is updated in real time by the system. 5. The updated status can be viewed on the monitor display by the security staff.
Alternative Flow	If a duplicate scan is detected within the cooldown period, the second scan is ignored by the system to prevent false toggling. If an entry after 8:00 PM without an approved late entry request is detected, Detect Unapproved Late Entry is triggered by the system.

Table 10: Use-Case Description 7

Field	**Description**
Use Case ID	UC008
Use Case Name	Detect Unapproved Late Entry
Description	Log Entry/Exit is extended by this case. This case is triggered automatically when a student entering the hostel after 8:00 PM (or before 6:00 AM) without an approved late access request is detected by the system.
Actor	System (automatic detection)
Pre-Condition	A student entry outside permitted hours (before 6:00 AM or after 8:00 PM) must have been logged. An approved late entry request for this date/time must not be possessed by the student.

Post-Condition	The late entry is flagged and recorded. The warden is notified of the unapproved late entry.
Main Flow	1. The entry timestamp falling outside permitted hours (6:00 AM – 8:00 PM) is detected by the system. 2. Whether an approved late access request for this date/time is held by the student is checked by the system. 3. If no approved request is found, the entry is flagged as an unapproved late entry by the system. 4. An alert notification is generated and sent to the warden by the system. 5. The incident is logged in the student's attendance history by the system.
Alternative Flow	If an approved late entry request is held by the student, the entry is recorded as normal and no alert is generated. If multiple students enter late simultaneously, each is processed and flagged individually by the system.

Table 11: Use-Case Description 8

Field	**Description**
Use Case ID	UC009
Use Case Name	Record Leave Status
Description	How a student is marked as "Going Home" through the mobile application during normal hours is described by this case. This is a self-service action by which warden approval is not required and no return date is needed.
Actor	Student
Pre-Condition	The mobile application must be logged into by the student. An active hostel account must be possessed by the student.
Post-Condition	The student's status is marked as "Going Home" in the system. The student is recorded as OUT on the attendance dashboard. A log entry is created in the leave history.
Main Flow	1. The leave management section in the mobile app is navigated to by the student. 2. The "Go Home" option is selected by the student. 3. A confirmation prompt is displayed by the system. 4. The action is confirmed by the student. 5. The leave is recorded with the current timestamp and the student is

	marked as OUT by the system. 6. The attendance dashboard is updated accordingly by the system. 7. When the student returns, their status is updated automatically upon scanning their ID card at the gate.
Alternative Flow	If the action is cancelled before confirmation by the student, no changes are made. If the student is already marked as OUT, the student is informed accordingly by the system.

Table 12: Use-Case Description 9

Field	**Description**
Use Case ID	UC010
Use Case Name	Request Late Entry/Exit
Description	How a request for late entry or exit outside the permitted hours (before 6:00 AM or after 8:00 PM) is submitted by a student is described in this case. The request is sent to the warden for approval.
Actor	Student
Pre-Condition	The mobile application must be logged into by the student. An active hostel account must be possessed by the student.
Post-Condition	The late access request is submitted with a "Pending" status. The warden is notified of the pending request.
Main Flow	1. The leave management section in the mobile app is navigated to by the student. 2. "Request Late Entry/Exit" is selected by the student. 3. A form requesting the expected date, time, and reason is displayed by the system. 4. The details are filled in and the request is submitted by the student. 5. The request is saved with a "Pending" status by the system. 6. A notification about the new request is received by the warden. 7. The request status can be tracked in the app by the student.
Alternative Flow	If mandatory fields are left empty, the student is prompted to complete them by the system. If a request for a past date/time is submitted, it is rejected automatically by the system. The request is processed by the warden through Approve/Reject Late Access Request.

Table 13: Use-Case Description 10

Field	**Description**
Use Case ID	UC011
Use Case Name	Approve/Reject Late Access Request
Description	Request Late Entry/Exit is extended by this case. How a student's late access request is reviewed and either approved or rejected by the warden is described.
Actor	Warden
Pre-Condition	The system must be logged into by the warden. At least one pending late access request must exist.
Post-Condition	The request status is updated to "Approved" or "Rejected." The student is notified of the decision.
Main Flow	1. The late access request section on the dashboard is navigated to by the warden. 2. All pending requests with student details, date/time, and reason are displayed by the system. 3. A request is selected to be reviewed by the warden. 4. The request is approved or rejected and comments are optionally added by the warden. 5. The request status is updated by the system. 6. A notification with the decision is sent to the student by the system. 7. If approved, the student's late entry/exit is permitted without an alert being triggered by the system.
Alternative Flow	If a response is not provided within a set time by the warden, a reminder notification is sent by the system. If the request is rejected by the warden, the student is notified with the reason provided.

Table 14: Use-Case Description 11

Field	**Description**
Use Case ID	UC012
Use Case Name	Submit Complaint
Description	Maintenance or disciplinary complaints are allowed to be submitted through the mobile application by students through this case. Complaints

	are text-based by default, with an optional image attachment feature.
Actor	Student
Pre-Condition	The mobile application must be logged into by the student. An active hostel account must be possessed by the student.
Post-Condition	The complaint is recorded with a "Pending" status and a unique complaint ID. The warden and sub-warden are notified.
Main Flow	1. The complaint section in the mobile app is navigated to by the student. 2. "Submit New Complaint" is selected by the student. 3. A form with fields for complaint category and description is displayed by the system. 4. The details are filled in and the complaint is submitted by the student. 5. The complaint is saved with a "Pending" status and a unique complaint ID is generated by the system. 6. The warden and sub-warden are notified of the new complaint by the system. 7. The status can be tracked in the complaint history by the student.
Alternative Flow	If mandatory fields are left empty, the student is prompted to complete them by the system. If images are wished to be attached by the student, Attach Images is invoked by the system. If submission fails due to a network error, the student is notified to retry by the system.

Table 15: Use-Case Description 12

Field	**Description**
Use Case ID	UC013
Use Case Name	Attach Images
Description	Submit Complaint is extended by this case. Images are allowed to be optionally attached as evidence when a complaint is being submitted by a student.
Actor	Student
Pre-Condition	The student must be in the process of submitting a complaint. Images must be available on the student's device.
Post-Condition	The selected images are uploaded and linked to the complaint record.

Main Flow	1. While the complaint form is being filled out, "Attach Image" is selected by the student. 2. The device's file picker or camera option is opened by the system. 3. An image is selected or captured by the student. 4. The image format and size are validated by the system. 5. The image is uploaded and attached to the complaint record by the system. 6. The attached image can be previewed before final submission by the student.
Alternative Flow	If the maximum file size is exceeded by the image, a smaller image is prompted to be selected by the system. If the image format is unsupported, an error is displayed and a valid format is requested by the system. An attached image may be removed before submission by the student.

Table 16: Use-Case Description 13

Field	**Description**
Use Case ID	UC014
Use Case Name	Receive Notifications
Description	How in-app notifications for various system events are received by students in the notification centre (inbox) is described by this case. Events including leave request decisions, announcements, complaint updates, and late entry warnings are covered.
Actor	Student
Pre-Condition	An active account must be possessed by the student, and the student must be logged in to the mobile app.
Post-Condition	The notification is stored in the student's in-app notification centre (inbox). Push notifications are not used.
Main Flow	1. A system event is triggered (e.g., leave approved/rejected, new announcement, complaint status update). 2. A notification with the relevant message is generated by the system. 3. The notification is stored for the student as unread in the in-app notification centre by the system. 4. The notification is listed with an unread indicator when the notification centre is opened by the student. 5. Notification details can be viewed by the student by tapping on it, and it is marked as read.

Alternative Flow	If the student is not using the app, the notification remains in the notification centre until the app is opened. Push notifications are not supported; notifications can only be viewed within the app.

Table 17: Use-Case Description 14

Field	**Description**
Use Case ID	UC015
Use Case Name	Monitor & View Attendance Dashboard
Description	A real-time dashboard displaying the current attendance status of all students is allowed to be viewed by the warden and sub-warden through this case. Counts of students inside, outside, and on leave are included.
Actor	Warden, Sub-Warden
Pre-Condition	The system must be logged into by the warden or sub-warden.
Post-Condition	The real-time attendance information is displayed to the user.
Main Flow	1. The system is logged into and the attendance dashboard is navigated to by the warden/sub-warden. 2. Real-time attendance data is retrieved from the database by the system. 3. Total counts of students currently IN, OUT, and on leave are displayed on the dashboard by the system. 4. Data can be filtered by room, floor, or batch by the user. 5. Individual students can be clicked on to view their detailed attendance history by the user. 6. The dashboard is auto-refreshed to reflect new scan events in real time by the system.
Alternative Flow	If no attendance data exists, a message indicating no records is displayed by the system. If real-time sync is temporarily delayed, the last known status with a timestamp is shown by the system.

Table 18: Use-Case Description 15

Field	**Description**

Use Case ID	UC016
Use Case Name	Resolve Maintenance Issues
Description	Maintenance complaints submitted by students are allowed to be viewed, managed, and resolved by the sub-warden through this case. Complaint progress is tracked and issues are marked as resolved once addressed.
Actor	Sub-Warden
Pre-Condition	The system must be logged into by the sub-warden. At least one complaint must exist in the system.
Post-Condition	The complaint status is updated. The student is notified when their complaint is resolved.
Main Flow	1. The complaint management section is navigated to by the sub-warden. 2. All complaints with status indicators (Pending, In Progress, Resolved) are displayed by the system. 3. Complaints can be filtered by status, date, or category by the sub-warden. 4. A complaint is selected and its details and any attached images are reviewed by the sub-warden. 5. The status is updated (e.g., marked as In Progress or Resolved) by the sub-warden. 6. The update is saved and the student is notified by the system.
Alternative Flow	If no complaints match the selected filter, "No complaints found" is displayed by the system. If a complaint is accidentally marked as resolved by the sub-warden, the status can be reverted.

Table 19: Use-Case Description 16

Field	**Description**
Use Case ID	UC017
Use Case Name	Generate Analytics & Reports
Description	Analytical reports on attendance patterns, leave statistics, complaint summaries, and room occupancy are allowed to be generated by the warden for administrative and decision-making purposes through this case.
Actor	Warden

Pre-Condition	The system must be logged into by the warden. Sufficient data must exist in the system for meaningful reports to be generated.
Post-Condition	The requested report is generated and displayed. The report can be exported or downloaded by the warden.
Main Flow	1. The analytics and reports module is navigated to by the warden. 2. Available report types (attendance, leave, complaints, occupancy) are displayed by the system. 3. A report type is selected and the date range and filters are specified by the warden. 4. The data is processed and the report with charts and summaries is generated by the system. 5. The report is reviewed on the dashboard by the warden. 6. The warden can export the report as PDF or CSV.
Alternative Flow	If insufficient data exists for the selected period, the warden is notified by the system. If report generation takes too long, a loading indicator is displayed by the system.

Table 20: Use-Case Description 17

Field	**Description**
Use Case ID	UC018
Use Case Name	Allocate Room/Bed
Description	Students are allowed to be assigned to specific rooms and beds, existing allocations are allowed to be updated, or allocations are allowed to be revoked by the warden through this case. Room assignments can be made during initial intake or as transfers at any time during the year.
Actor	Warden
Pre-Condition	The system must be logged into by the warden. An active account must be possessed by the student. Available capacity must exist in the target room/bed (for new assignments).
Post-Condition	The student is assigned to the selected room and bed. The room availability is updated in the system.
Main Flow	1. The room allocation module is navigated to by the warden. 2. All rooms with current occupancy status and available beds are displayed by the system. 3. A room and an available bed are selected by the warden.

	4. The student to be assigned is searched for and selected by the warden. 5. The bed availability and room capacity are validated by the system. 6. The student is assigned to the bed and allocation records are updated by the system. 7. A confirmation message is displayed by the system.
Alternative Flow	If the selected bed is already occupied, the assignment is blocked and the warden is notified by the system. If an existing allocation is already held by the student, a transfer confirmation is requested from the warden by the system (the old bed is freed). If an allocation is revoked by the warden, the bed is marked as available by the system.

Table 21: Use-Case Description 18

Field	**Description**
Use Case ID	UC019
Use Case Name	Monitor System Display
Description	How the system status display at the hostel gate is physically monitored by security staff is described by this case. Scan results and student verification status are shown in real time on the display, allowing the entry/exit process to be observed by security.
Actor	Security (Physical Monitor)
Pre-Condition	The system display/monitor at the gate must be operational. The ESP32 device must be connected and functioning.
Post-Condition	Visual confirmation of student entry/exit events as they occur is obtained by the security staff.
Main Flow	1. The hostel gate near the system display is stationed by the security staff. 2. When the ID card is scanned by a student, the scan result (verified/rejected, student name, IN/OUT status) is shown on the display by the system. 3. The display is visually monitored for any rejected or suspicious scans by the security staff. 4. If a rejected scan is displayed, physical intervention may be performed by the security staff. 5. The display is reset and the next scan event is awaited by the system.

Alternative Flow	If the display malfunctions, physical ID card verification and manual records are relied upon by the security staff. If a rejected scan is shown, the student may be asked to re-scan or identity is verified manually by the security staff.

Table 22: Use-Case Description 19

Field	**Description**
Use Case ID	UC020
Use Case Name	Maintain Physical Records
Description	The fallback process where student entry and exit information is manually recorded in a physical logbook by security staff when the automated system is unavailable or as a supplementary backup is described by this case.
Actor	Security (Physical Monitor)
Pre-Condition	The hostel gate must be stationed at by the security staff. A physical logbook must be available.
Post-Condition	The student's entry/exit is recorded in the physical logbook. Manual records can later be reconciled with the digital system.
Main Flow	1. The hostel gate is approached by a student. 2. The university ID card is requested to be presented by the security staff. 3. The student's identity is visually verified by the security staff. 4. The student's name, ID number, time, and entry/exit direction are recorded in the physical logbook by the security staff. 5. The student is permitted to enter or leave.
Alternative Flow	If the automated system is restored, manual recording may be discontinued and the digital system is relied upon by the security staff. If the ID card is not possessed by the student, the incident is recorded and entry may be denied by the security staff.

Table 23: Use-Case Description 20

4.3 Activity Diagrams

[Figure 2: Activity Diagram – Smart QR Access and Security Workflow (ESP32 Integration)]

[Figure 3: Activity Diagram – Late Access Request and Approval Lifecycle]

[Figure 4: Activity Diagram – Leave Recording and Log Workflow]

[Figure 5: Activity Diagram – Complaint Tracking and Maintenance Resolution Workflow]

[Figure 6: Activity Diagram – System Downtime and Manual Fallback Synchronization]

[Figure 7: Activity Diagram – Student Onboarding, Room Allocation & Setup Workflow]

[Figure 8: Activity Diagram – Announcement & Notice Broadcast Workflow]

CHAPTER 5

PROJECT ANALYSIS & FEASIBILITY

5.1 Alternative Solutions Considered

During the development of the HostelHive system, several alternative solutions were considered to address the problems associated with the existing manual hostel management system. Each alternative was evaluated based on efficiency, cost, scalability, and ease of implementation. The selected solution was chosen after comparing these alternatives.

5.1.1 Manual Paper-Based System

The existing system is based on manual record-keeping using paper logbooks. Student entry and exit, room allocation, leave requests, and complaint management are all recorded manually.

Advantages: Low initial cost; No technical knowledge required

Disadvantages: Time-consuming processes; High risk of data loss and errors; Difficult to track student movements; Lack of real-time monitoring

This approach was not selected due to inefficiency and lack of scalability.

5.1.2 Basic Digital System (Without QR Integration)

A simple digital system using spreadsheets or basic web applications was considered. In this approach, hostel data would be digitized, but entry and exit tracking would still rely on manual input.

Advantages: Reduced paperwork; Easier data storage and retrieval

Disadvantages: Manual errors may still occur; No real-time tracking of student movements; Limited security features; Lack of automation

This solution was not selected as it does not fully address the problem of real-time monitoring and security.

5.1.3 Biometric-Based System

A biometric system (fingerprint or facial recognition) was considered for tracking student attendance and entry/exit.

Advantages: High accuracy in identifying students; Improved security

Disadvantages: High implementation and maintenance cost; Privacy concerns; Hardware dependency and complexity; Possible failures due to hardware issues

This option was not selected due to high cost and complexity compared to the project scope.

5.1.4 RFID-Based System

An RFID-based solution using smart cards was evaluated for entry and exit tracking.

Advantages: Faster scanning process; Contactless operation

Disadvantages: Additional hardware cost; Risk of card loss or misuse; Requires dedicated infrastructure

This solution was not selected due to additional cost and hardware requirements.

5.1.5 Proposed Solution: HostelHive System (QR-Based System)

The HostelHive system, in which QR code-based tracking is integrated with a digital management platform, was selected as the final solution.

Advantages: A cost-effective approach is provided compared to biometric and RFID systems; Easy implementation and maintenance are supported; Real-time tracking of student movements is facilitated; Manual errors and paperwork are reduced; Additional features such as leave management, complaints, and announcements are provided.

Disadvantages: The possibility of QR code misuse (e.g., duplication) is presented. (This risk is acknowledged but considered outside the current system scope and can be addressed in future enhancements.)

Conclusion of Alternatives
After evaluating all alternatives, the QR-based HostelHive system was selected as the most suitable solution due to its balance of cost, functionality, ease of use, and scalability. It effectively addresses the limitations of the existing manual system while providing a foundation for future improvements.

5.2 Feasibility Study

A feasibility study was conducted to evaluate the practicality of developing and implementing the HostelHive system. The study examines the project from technical, economic, operational, and schedule perspectives to ensure that the proposed solution is viable and achievable.

5.2.1 Technical Feasibility

Whether the required technology, tools, and resources are available for the system to be developed and operated is evaluated by technical feasibility.

- The system shall be developed using modern and widely supported technologies such as Flutter and React for frontend development, Go for backend services, and PostgreSQL for database management.
- QR code scanning shall be implemented using an ESP32 microcontroller, which is cost-effective and suitable for embedded systems.
- The required development tools, frameworks, and libraries are readily available and well-documented.
- The necessary technical knowledge for the system to be implemented is possessed by the development team.
- Integration between hardware (ESP32) and software components shall be achieved using standard communication protocols.
- Therefore, the system is considered technically feasible.

5.2.2 Economic Feasibility

Whether the project is financially viable and cost-effective is evaluated by economic feasibility.

- Open-source technologies are utilized by the system, by which software costs are significantly reduced.
- The hardware component (ESP32) is low-cost and affordable compared to alternatives such as biometric or RFID systems.
- Long-term cost savings are achieved through the reduction of paper usage and manual processes.
- Maintenance and operational costs are expected to be minimal.

Therefore, the system is considered economically feasible.

5.2.3 Operational Feasibility

How well the system fits within the existing operational environment and whether it can be effectively used by users is evaluated by operational feasibility.

- The system is designed so that manual processes are replaced with a simple and user-friendly interface.
- Minimal training is required for users such as students, wardens, and security staff.
- Efficiency is improved by the system through the automation of routine tasks such as attendance tracking and leave management.
- Hostel security and management are enhanced by real-time monitoring.
- The operational needs of Bulugaha Girls' Hostel are aligned with by the system.

Therefore, the system is considered operationally feasible.

5.2.4 Schedule Feasibility

Whether the project can be completed within the given time constraints is evaluated by schedule feasibility.

- The project is divided into manageable phases, including requirement analysis, design, development, testing, and deployment.
- A modular development approach is adopted, by which the parallel development of system components is facilitated.
- Rapid development and implementation are supported by the selected technologies.
- The project scope is clearly defined so that unnecessary delays are avoided.
- The system is expected to be completed within the allocated academic project timeline.

Therefore, the system is considered schedule feasible.

5.3 Risk Analysis

Potential risks associated with the development and deployment of the HostelHive system are identified, categorized, and evaluated in this section. Mitigation strategies are proposed so that the impact and likelihood of these risks are minimized.

Risk ID	Description	Category	Likelihood	Impact	Risk Level	Mitigation Strategy
R001	Potential integration issues between the ESP32	Technical	Medium	High	High	Standardized APIs and rigorous integration

	hardware and the backend API.					testing shall be utilized.
R002	Risk of QR code duplication or unauthorized misuse by students.	Technical	Medium	Medium	Medium	Time-sensitive dynamic QR codes and additional validation logic shall be implemented.
R003	System performance degradation under high concurrent load during peak hours.	Technical	Low	High	Medium	Load testing and database query optimization shall be performed.
R004	Delays in project development phases due to unforeseen technical complexities.	Schedule	Medium	High	High	A buffer shall be included in the project timeline and milestones shall be monitored weekly.
R005	Limited development time due to strict academic deadlines.	Schedule	High	Medium	High	A Minimum Viable Product (MVP) shall be prioritized to ensure core features are delivered first.
R006	Lack of advanced technical expertise in C++ for microcontroller programming.	Resource	Medium	Medium	Medium	Available technical documentation and peer-led learning resources shall be utilized.

R007	Unavailability or shipping delays of required hardware components (ESP32, QR module).	Resource	Low	High	Medium	Critical hardware shall be procured at the earliest possible stage of the project.
R008	Changes in system requirements by hostel administration during the development phase.	Requirements	Medium	High	High	A formal change control process shall be established and requirements shall be finalized early.
R009	Incomplete or ambiguous requirements provided by stakeholders.	Requirements	Medium	Medium	Medium	Regular communication and requirement validation meetings shall be maintained.
R010	Unstable network connectivity at the hostel gate by which real-time updates are affected.	External	Medium	Medium	Medium	Offline data caching and retry mechanisms shall be implemented on the ESP32.
R011	Potential data security breaches or unauthorized access to sensitive student data.	External	Low	High	Medium	Industry-standard encryption and robust authentication protocols shall be enforced.
R012	Resistance to change from administrative users	Organizational	Medium	Medium	Medium	Comprehensive user training and awareness sessions shall

	associated with the transition from manual logbooks.					be provided to the staff.

Table 24: Risk Analysis

CHAPTER 6

REFERENCES

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7. Client Requirement Sign-Off Page

The undersigned parties confirm that the scope, functional requirements, non-functional requirements, and features documented in this Software Requirement Specification have been reviewed, discussed, and approved. This document will serve as the basis for the design and development of the system.

Without client approval, the project will not proceed to the design phase.

□ Scope Approved
□ Requirements Approved
□ Features Approved

Client Representative Sign-Off
Name: _____
Title: _____
Organisation: _____
Date: _____
Signature: _____

Project Supervisor Sign-Off
Name: _____
Date: _____
Signature: _____